

Activity 1: Anticipating Sociotechnical Impact in Control Systems Design

Control Systems Course - 2026-1
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1 Sociotechnical Mapping of a Controlled Two-Link Walker System

1.1 Course Context

This activity is implemented in an introductory Control Systems course for undergraduate Electronic Engineering students. The technical project consists of modelling, simulating, and controlling a simplified *two-link one-leg walker*, inspired by rehabilitation-oriented robotic systems such as pediatric exoskeletons and assistive gait devices.

The purpose of the activity is to help students understand that a control system is not only defined by its mathematical model or controller performance, but also by the way it interacts with users, clinicians, caregivers, institutions, and real-world environments.

1.2 Motivation

In control engineering, systems are usually evaluated through technical criteria such as stability, tracking error, settling time, overshoot, energy consumption, and robustness. These criteria are essential, but they are not sufficient when the system is intended for human-centered applications such as rehabilitation technologies.

A controller may be mathematically stable and experimentally successful in the laboratory, but still fail in practice if it is unsafe, difficult to operate, inaccessible, too expensive, poorly adapted to clinical workflows, or unsuitable for the needs of patients and therapists.

Therefore, this activity introduces students to **sociotechnical thinking** through a simplified Responsible Innovation approach. Students are expected

to anticipate how their engineering design could behave in a real context and how this analysis should influence control design decisions.

1.3 Learning Outcomes

At the end of this activity, students will be able to:

1. Contextualize a control system within a real-world rehabilitation-inspired application.
2. Identify stakeholders and operational environments associated with a controlled robotic system.
3. Anticipate potential benefits, risks, limitations, and unintended consequences of the system.
4. Translate sociotechnical insights into concrete engineering design decisions.

This activity is aligned with:

- Each task is connected to an observable and assessable learning outcome.
- Engineering design under realistic constraints, including health, safety, welfare, societal, and environmental considerations.
- The activity emphasizes the transition from *Conceive* to *Design*, encouraging students to understand the system context before proposing technical solutions.

1.4 Core Concept: Responsible Innovation

Responsible Innovation is an engineering and research approach that promotes the design of systems that are not only technically valid, but also safe, usable, socially relevant, and appropriate for real contexts.

In this activity, Responsible Innovation is introduced through the action of **anticipation**. Anticipation means thinking ahead about what could happen when a technology is used outside the laboratory.

Students will use four guiding lenses:

- **People:** Who uses or is affected by the system?
- **Product:** Is the system safe, usable, reliable, and accessible?
- **Process:** What failures, risks, or limitations could emerge?
- **Purpose:** Why does this system exist, and who benefits from it?

1.5 Technical Control-System Framing

The two-link walker is modeled as a simplified planar mechanical system with two generalized coordinates:

$$q = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}, \quad \dot{q} = \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \end{bmatrix}, \quad \ddot{q} = \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{\theta}_2 \end{bmatrix}.$$

The nonlinear dynamics can be represented in standard robotic form as:

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau + d(t),$$

where:

- $M(q)$ is the inertia matrix,
- $C(q, \dot{q})$ represents Coriolis and centrifugal effects,
- $G(q)$ represents gravitational torques,
- τ is the control input,
- $d(t)$ represents disturbances or unmodeled effects.

For control design, students may use a linearized approximation around an operating point:

$$x = \begin{bmatrix} q \\ \dot{q} \end{bmatrix}, \quad \dot{x} = Ax + Bu, \quad y = Cx,$$

where x is the state vector, u is the control input, and y is the measured output.

A possible state-feedback controller is:

$$u(t) = -Kx(t) + r(t),$$

where K is the feedback gain and $r(t)$ is a reference or feedforward term.

The technical objective may be expressed as achieving stability and reference tracking:

$$\lim_{t \rightarrow \infty} e(t) = 0, \quad e(t) = q_{\text{ref}}(t) - q(t).$$

However, in a sociotechnical context, the design objective must also include safety and usability constraints. For example:

$$|\tau_i(t)| \leq \tau_{i,\text{max}}, \quad |\theta_i(t)| \leq \theta_{i,\text{max}}, \quad |\dot{\theta}_i(t)| \leq \dot{\theta}_{i,\text{max}}.$$

Thus, the control problem is not only:

Minimize tracking error,

but rather:

Minimize tracking error while satisfying safety, usability, and contextual constraints.

A simplified performance index can be written as:

$$J = \int_0^T (e(t)^\top Q e(t) + u(t)^\top R u(t)) dt,$$

subject to:

$$|\tau_i(t)| \leq \tau_{i,\max}, \quad |\theta_i(t)| \leq \theta_{i,\max}, \quad \text{safe human interaction.}$$

This formulation helps students understand that sociotechnical thinking can influence the mathematical and engineering definition of the control problem.

1.6 Activity Pipeline

1.7 Learning Activity

1.7.1 Step 1: Technical System Description

Each group must provide a concise technical description of the two-link walker system. The description should include:

- The dynamic model or simplified representation used.
- The state variables and control inputs.
- The control strategy selected.
- The main technical objective of the system.

Students should explicitly state whether their control objective is related to stabilization, trajectory tracking, disturbance rejection, energy reduction, or another control goal.

1.7.2 Step 2: Stakeholder and Context Mapping

Students must identify the people and contexts that would be involved if a walker-inspired control system were translated into a rehabilitation technology.

Each group must include at least:

- one direct user,
- one professional actor,
- one indirect user,
- one environmental or institutional context.

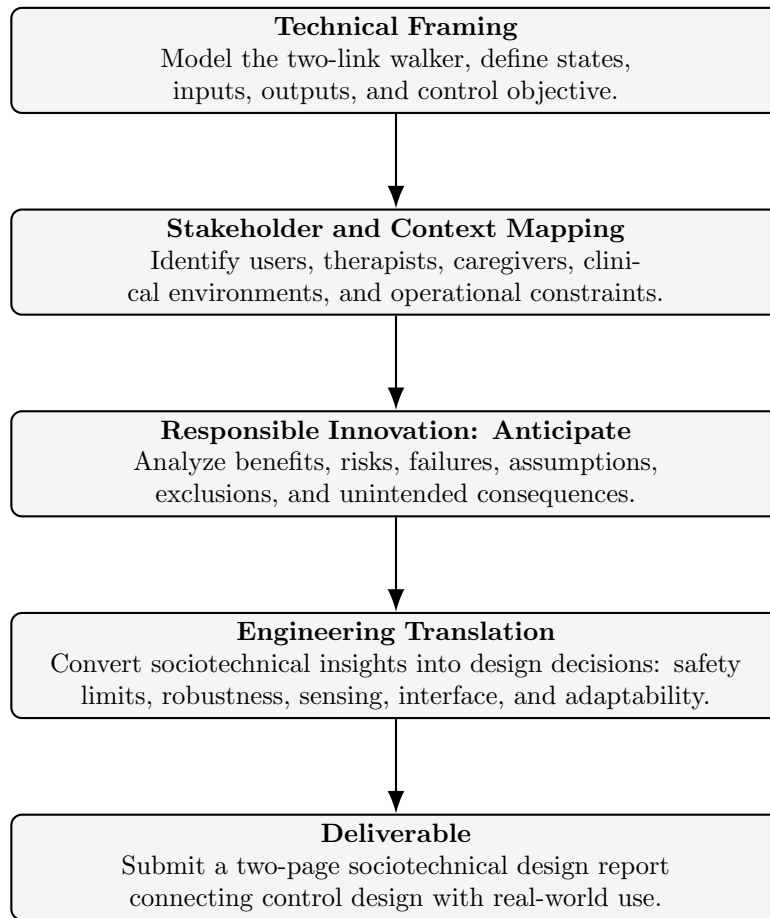


Figure 1: Pipeline of Activity 1: from technical modelling to sociotechnical design decisions.

1.7.3 Step 3: Anticipation of Impacts and Risks

Using the Responsible Innovation lens of **Anticipation**, students must answer the following questions:

1. What benefits could this system provide in a rehabilitation-inspired context?
2. What technical limitations could affect real-world performance?
3. What could fail if the system is used outside the laboratory?
4. What assumptions are being made about the user, therapist, or environment?

Table 1: Stakeholder and context mapping template.

Stakeholder Context	or	Type		Interaction with the System
Child or patient		Direct user		Physically interacts with the assistive system.
Physiotherapist		Professional user		Supervises, adjusts, and interprets system behavior.
Caregiver or family		Indirect user		Supports therapy continuity and adoption.
Clinic or rehabilitation center		Operational context	con-	Provides space, protocols, and institutional constraints.
Home environment		Deployment context	con-	Introduces variability, accessibility constraints, and safety issues.

5. Who might be excluded or disadvantaged by this design?

6. What unintended consequences could emerge from the system?

Students are expected to go beyond listing issues. They must explain cause-effect relationships.

For example:

If the controller assumes a fixed user mass and stiffness, then performance may degrade when interacting with children of different sizes or motor conditions. This may require adaptive control, gain scheduling, or online parameter estimation.

1.7.4 Step 4: Engineering Design Implications

Students must translate the previous analysis into concrete engineering decisions.

Each group must propose at least three design modifications. Each modification must include:

- the sociotechnical issue identified,
- the engineering decision proposed,
- the technical justification,
- the expected benefit for real-world use.

Table 2: Template for translating sociotechnical analysis into engineering decisions.

Issue Identified	Design Decision	Justification
Excessive torque may harm the user.	Add torque saturation limits.	Safety constraints reduce the probability of unsafe actuation.
User variability affects model accuracy.	Use robust or adaptive control.	The controller becomes less sensitive to parameter uncertainty.
Therapists may not understand complex parameters.	Simplify the user interface.	Improves usability and adoption in clinical contexts.
Sensor noise may affect feedback.	Add filtering or sensor redundancy.	Improves reliability of measured states.

1.8 Deliverable

Each group must submit a technical report with a maximum length of two pages.

The report must include:

1. **Technical system description:** model, states, inputs, and control objective.
2. **Stakeholder and context mapping:** users, professionals, institutions, and environments.
3. **Impact and risk analysis:** benefits, risks, assumptions, exclusions, and unintended consequences.
4. **Design implications:** at least three justified engineering design decisions.

1.9 Assessment Rubric

1.10 Grading Weights

1.11 Expected Learning Closure

At the end of this activity, students should recognize that:

A control system is not only defined by its equations, poles, gains, or simulation results. It is also defined by how it interacts with people, environments, risks, constraints, and institutions.

This is especially important in health, rehabilitation, robotics, and intelligent systems, where technical performance must be integrated with safety, usability, accessibility, and real-world adoption.

Table 3: Assessment rubric for Activity 1.

Criterion	Low Level	Medium Level	High Level
Technical clarity	The system description is vague or incomplete.	The system is partially described, but some variables or objectives are unclear.	The system is clearly described, including model, variables, inputs, and control objective.
Stakeholder mapping	Stakeholders are generic or incomplete.	Main stakeholders are identified, but contextual relationships are weak.	Stakeholders and contexts are complete, specific, and connected to system use.
Impact and risk analysis	The analysis is mostly descriptive and lists issues without explanation.	The analysis includes some reasoning about benefits and risks.	The analysis is critical, contextualized, and explains cause-effect relationships.
Engineering translation	Design implications are absent or disconnected from the analysis.	Some design decisions are proposed, but justification is limited.	Design decisions are concrete, technically justified, and directly derived from sociotechnical analysis.

Table 4: Grading weights for Activity 1.

Component	Weight
Technical system description	20%
Stakeholder and context mapping	25%
Impact and risk anticipation	25%
Engineering design implications	30%